

One-line: What makes employees recommend their employer (or not), and where HR should act first for the biggest return.

29%

Would NOT
recommend

£1.15–2.69M

Annual cost
exposure /1,000

2.46

Culture odds
ratio (top driver)

0.86

Model recall
on detractors

30%

Flagged
High-risk

549K

Analysis-ready
reviews

Business Context

- Low advocacy is a financial problem, not just reputational: Gallup puts the cost of one disengaged employee at ~34% of salary.
- The dataset has no attrition or salary columns, so the 'recommend' field is used as an honest PROXY for advocacy and retention intent.
- Findings describe controlled association, not proven causation. Cost figures are illustrative exposure under stated assumptions, not guaranteed ROI.

The Four Business Questions

Q1 — WHY do employees not recommend? Which sub-ratings separate promoters from detractors? (Sections 6, 9)

Q2 — WHAT are employees actually saying, in their own words? (NLP / Module F)

Q3 — WHO is most at risk, and can we predict it? (Sections 8, 10)

Q4 — HOW should the organization act, and what is the payoff? (Sections 11, 12)

Driver Ranking — recommend-rate gap (low 1–2 vs high 4–5 rating)

Driver	Low rating recommends	High rating recommends	Gap (pts)	Rank
Culture & Values	17%	93%	77	1 (strongest)
Career Opportunities	24%	92%	68	2 (tied)
Senior Management	28%	96%	68	2 (tied)
Compensation	33%	88%	56	4
Work-Life Balance	38%	88%	50	5

The Compounding Effect (critical finding)

- Culture STRONG + Sr Mgmt STRONG: 98% recommend
- Culture WEAK + Sr Mgmt WEAK: 12% recommend

Weak-on-both collapses disproportionately → fix culture & leadership TOGETHER, not one at a time.
Verified on full data: smallest cell still >4,000 reviews (robust, not a fluke).

Chi-Square Validation

Driver	Chi-Square	p-value	Sig?
Culture & Values	250,614	<0.001	Yes
Senior Management	229,511	<0.001	Yes
Career Opportunities	200,306	<0.001	Yes
Compensation	128,566	<0.001	Yes
Work-Life Balance	113,856	<0.001	Yes

Two independent methods (gap + chi-square) agree: Culture leads. That agreement is what makes the ranking trustworthy.

Model Comparison — benchmark to justify simplicity

Model	Recall (detractor)	ROC-AUC	PR-AUC	Selected?
Logistic Regression (balanced)	0.86	0.937	0.969	YES — interpretable
Random Forest (benchmark)	—	0.938	0.971	No (~0.1% better only)

- Why Logistic wins: RF only edges it by ~0.1% → interpretable odds ratios (explainable to a CHRO) outweigh a fractional gain.
- Optimized for RECALL on 'not recommend': missing a detractor is costly; a false alarm only costs a conversation (cost asymmetry).
- Honest caveat: high scores partly reflect that sub-ratings and the recommend decision share the same underlying sentiment.

Risk Scoring — well-calibrated tiers (full data)

Risk band	Threshold	% of population	Actual detractor rate	Action
High	risk >= 0.60	30%	78%	Prioritize — engage first
Medium	0.30–0.60	15%	26%	Lighter-touch monitoring
Low	risk < 0.30	56%	3.5%	No active intervention

- Coverage: the High band holds 30% of people but 78% of them are true detractors — HR effort lands on real risk, not noise.
- Trade-off (deliberate): ~22% of the High band are false alarms; accepted because missing a detractor costs far more.
- Former employees concentrate risk (39.6% High vs 23.2% current) but have left → target by weak DRIVER, not demographic.

Intervention Priority — three criteria: driver strength × capital efficiency × time-to-impact

Priority	Intervention	Targets driver	Impact	Effort / time
P1	Leadership & Culture program	Culture + Sr Mgmt (2.46 / 2.27)	Highest	High effort, slow 6–12mo
P2	Career Pathway framework	Career Opportunities (2.19)	High	Medium, structural
P3	Grey-zone monitoring	Medium-risk drift	Preventive	Low, fast

Impact-over-efficiency choice, stated openly. 'Benchmark first': pay is a SECONDARY driver, so no blanket pay rises.

Cost-Benefit (illustrative exposure)

- 1,000-person org → ~290 high-risk employees
- Annual exposure: ~£1.15M (Gallup 34%-of-salary)
- Effectiveness shown as 15/30/45% brackets —
- NOT empirical; true effect needs the pilot.
- Gallup: managers ≈ 70% of engagement variance.

Voice of Employee (NLP / n-grams)

- Detractors mention management 2.5x more than promoters.
- Promoters mention work conditions 2.6x more — yet still
- recommend → conditions tolerated, bad mgmt is the dealbreaker.
- Senior detractors: 'long term', 'non existent' (growth/direction).
- BERTopic failed on short text → n-grams (method fit > prestige).

KEY CAVEATS (say these out loud in interview — they are your credibility)

- Association, not causation — the model shows where to act; a randomized pilot (Section 13) proves it works.
- 'Recommend' is a PROXY for advocacy — no attrition/salary data exists. Conclusions are about advocacy, not turnover directly.
- High scores partly built-in: sub-ratings & recommend share the same sentiment. Anonymous, review-level, UK, 2008–2021.